

AI- Based Fake News Detection System

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Abstract—This paper presents the design and development of an AI-Based News Analysis and Credibility Verification System developed using React, Vite, FastAPI, and machine learning technologies [1]-[5]. The system is designed to analyze news headlines and articles by automatically detecting the news genre, collecting supporting online references, evaluating evidence credibility, and generating a final verdict regarding the authenticity of the submitted content. The application integrates multiple artificial intelligence models including SBERT semantic similarity analysis, Natural Language Inference (NLI), fake-news classification, sentiment analysis, and zero-shot credibility models to improve verification accuracy [3],[4],[6]-[13]. The frontend is developed using ReactJS with Vite and Tailwind CSS to provide a responsive and interactive user interface, while the backend is implemented using Python FastAPI and transformer-based machine learning models [1]-[5] for processing and analysis. The proposed system provides verdicts such as Likely True, Likely False, or Needs More Evidence along with confidence scores, evidence references, and credibility signals. The developed solution helps users identify misleading information, improves digital media awareness, and supports automated fact-checking processes.

Index Terms—Fake News Detection, News Analysis System, Artificial Intelligence, FastAPI, ReactJS, Machine Learning, SBERT, Natural Language Processing, Evidence Verification, Credibility Analysis.

I. INTRODUCTION

The rapid growth of social media platforms and online news portals has significantly increased the spread of misinformation and fake news across digital environments [6]-[13]. Millions of users consume and share online information daily, making it difficult to verify whether the content is authentic, misleading, or completely false. Traditional manual fact-checking methods are time-consuming and often unable to handle the large volume of

information generated on the internet [12],[13].

To address these challenges, the proposed “AI-Based News Analysis and Credibility Verification System” provides an intelligent web-based platform capable of analyzing news articles and verifying their credibility using artificial intelligence and machine learning techniques [6]-[13]. The system allows users to enter a news headline or article text through a frontend interface, after which the backend processes the content and performs automated analysis.

The system first cleans and preprocesses the input text before detecting the news genre such as Sports, Politics, Business, Entertainment, or General News. It then collects online references from Google News RSS feeds and Reddit public search APIs to compare the submitted claim with publicly available information. Evidence scoring mechanisms calculate semantic similarity and stance relationships between the claim and the collected references [3],[4].

The proposed system integrates multiple AI-based credibility signals including fake-news classification models, zero-shot credibility analysis, sentiment risk analysis, sensational language detection, claim specificity analysis, and trusted source evaluation [6]-[13]. Based on the combined evidence and model outputs, the system generates a final verdict such as Likely True, Likely False, or Needs More Evidence. The frontend of the application is developed using ReactJS, Vite, Tailwind CSS, and modern UI components to provide a responsive user experience. The backend is implemented using Python FastAPI framework with transformer-based NLP models and asynchronous APIs for scalable processing [2]-[5]. The developed system improves transparency, reduces manual fact-checking effort, and provides users with explainable AI-based news verification.

II. LITERATURE SURVEY

The increasing spread of misinformation and fake news has become a major challenge in digital communication systems. Several researchers and organizations have proposed machine learning and natural language processing techniques for automated fake news detection and credibility analysis [6]-[13].

Existing fake news detection systems generally rely on text classification models trained using supervised machine learning techniques. Traditional approaches mainly use keyword analysis, TF-IDF vectorization, sentiment analysis, and Naive Bayes or Support Vector Machine algorithms for identifying fake or misleading content. Although these methods provide moderate accuracy, they often fail to understand semantic relationships and contextual meaning within news articles.

Recent advancements in transformer-based deep learning models such as BERT, RoBERTa, and Sentence Transformers have significantly improved text understanding and semantic similarity analysis [3],[4],[6]-[13]. Several modern systems use semantic embeddings and Natural Language Inference models to compare claims with supporting evidence from external sources.

Many existing fact-checking platforms collect references from trusted online news websites and compare them with user-submitted claims. However, several systems still face limitations related to real-time evidence collection, multi-source verification, stance detection, and explainability of AI decisions.

Modern web technologies such as ReactJS, FastAPI, and asynchronous APIs are increasingly used for building scalable AI-powered web applications[1],[2]. FastAPI provides efficient backend API handling with high performance, while ReactJS enables responsive frontend interfaces and dynamic data visualization.

The proposed system addresses the limitations of existing fake news detection systems by integrating semantic similarity analysis, evidence-based verification, genre detection, stance analysis, model ensemble signals, and explainable AI outputs into a single unified platform [11],[12]. The system improves reliability, transparency, and scalability in automated news credibility verification.

Several recent studies have also focused on explainable artificial intelligence techniques for misinformation detection systems. Explainable AI approaches provide transparency by showing supporting evidence, confidence scores, and reasoning behind classification results. Research indicates that users are more likely to trust AI-generated decisions when systems provide interpretable outputs instead of black-box predictions. Modern misinformation detection platforms increasingly combine deep learning models with evidence-based reasoning and external source verification to improve both prediction accuracy and user confidence.

A. Comparative Analysis of Existing Systems

TABLE I
COMPARISON OF EXISTING SYSTEMS

Feature	Existing System	Proposed System
News Verification	Manual	Automated
Context Analysis	Limited	Advanced NLP-Based
Classification Accuracy	Moderate	High
Real-Time Detection	No	Yes
Semantic Understanding	Basic	Context-Aware

Existing fake news detection systems primarily rely on traditional machine learning techniques and basic text classification approaches. These systems often require manual feature extraction and are limited in their ability to understand the contextual meaning of news content. As a result, their classification accuracy is moderate, and they may fail to detect sophisticated misinformation that uses complex language patterns. Furthermore, most existing solutions do not support real-time analysis and lack user-friendly interfaces for effective interaction.

The proposed AI-Based Fake News Detection and Analysis System addresses these limitations by incorporating advanced Natural Language Processing (NLP) and transformer-based models. Unlike traditional approaches, the proposed system automatically extracts features from news articles and performs context-aware semantic analysis to better understand the meaning and credibility of the content. This significantly improves the accuracy and reliability of fake news detection.

Additionally, the proposed system supports real-time prediction, enabling users to verify news articles instantly. The integration of an interactive user dashboard enhances usability by providing clear classification results and analytical insights. The system is also highly scalable, allowing it to process large volumes of news data efficiently. Therefore, compared to existing methods, the proposed solution offers improved performance, higher accuracy, better contextual understanding, and a more efficient approach to combating misinformation.

Moreover, the proposed system is designed to adapt to the rapidly evolving nature of online misinformation. By leveraging advanced AI algorithms and continuously updating the training dataset, the system can identify emerging fake news patterns and misleading content more effectively. The combination of automated analysis, contextual understanding, and real-time processing not only enhances detection accuracy but also reduces human effort in news verification. Consequently, the proposed system serves as a reliable and efficient solution for assisting users in distinguishing between authentic and misleading news articles in the digital era.

III. SYSTEM ARCHITECTURE

The proposed AI-Based News Analysis and Credibility Verification System follows a centralized client-server architecture that integrates frontend technologies, backend services, database management, and secure authentication mechanisms into a unified platform [1]-[5]. The architecture is designed to support multiple user roles including Students, Faculty Members, Accountants, Exam Cell Staff, and Administrators through dedicated dashboards and controlled system access.

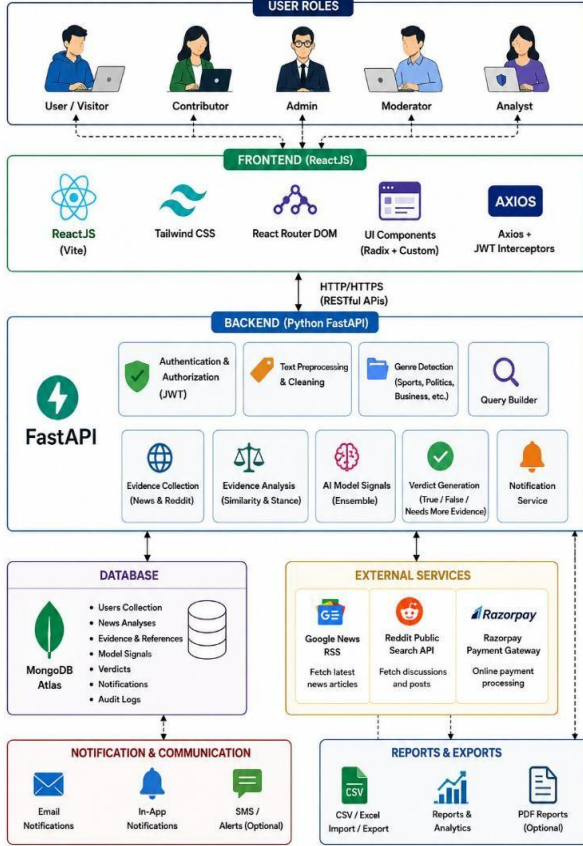


Fig. 1. System Architecture of Fake News Detection System

The proposed AI-Based News Analysis and Credibility Verification System follows a centralized client-server architecture that integrates frontend technologies, backend APIs, artificial intelligence models, and external evidence collection services into a unified platform. The architecture is designed to automate news credibility analysis by processing user-submitted news content, collecting supporting references, evaluating evidence, and generating explainable AI-based verdicts.

The architecture of the system consists of multiple interconnected modules including the Frontend User Interface, FastAPI Backend Server, Text Processing Module, Genre Detection Module, Evidence Collection Module, AI Model Signal Module, Verdict Generation Module, and External Reference APIs. The frontend layer is developed using ReactJS with Vite for faster rendering and efficient component management. Tailwind CSS is used to design a responsive and interactive user interface, while reusable UI components provide analysis panels, result cards, evidence displays,

and visualization sections. The frontend communicates with the backend using asynchronous API requests through the Vite proxy configuration.

The backend layer is implemented using Python FastAPI framework, which provides RESTful APIs for handling request validation, text processing, evidence collection, AI model execution, and final verdict generation[2],[5]. FastAPI also supports asynchronous processing to improve system performance and scalability during multiple evidence retrieval operations.

The text preprocessing module cleans and normalizes user-submitted news content by removing unnecessary symbols, HTML tags, and extra whitespace. After preprocessing, the Genre Detection Module identifies the category of news such as Sports, Politics, Business, Entertainment, or General News using keyword-based scoring and contextual analysis techniques [3],[4],[6]-[13].

The Evidence Collection Module gathers related references from external online sources including Google News RSS feeds and Reddit public search APIs. The collected references are normalized into structured formats and forwarded to the evidence analysis module for comparison with the submitted claim. The modular architecture ensures scalability, maintainability, transparency, and efficient processing for automated news verification systems while supporting future enhancements such as multilingual analysis, image-based fact-checking, and cloud deployment.

IV. MATHEMATICAL MODEL

The proposed AI-Based Fake News Detection and Analysis System can be represented mathematically using set theory, feature extraction functions, classification models, and performance evaluation metrics. The mathematical model defines system entities, news classification procedures, prediction mechanisms, and system accuracy calculations.

A. Set Theory Representation

Let

$$\begin{aligned}
 N &= \{\text{News Articles}\} \\
 U &= \{\text{Users}\} \\
 P &= \{\text{Preprocessed News Data}\} \\
 F &= \{\text{Extracted Features}\} \\
 M &= \{\text{Machine Learning Model}\} \\
 C &= \{\text{Classifications}\} \\
 R &= \{\text{Results}\} \\
 D &= \{\text{Database}\}
 \end{aligned}$$

The complete Fake News Detection System can be represented as:

$$S = \{N, U, P, F, M, C, R, D\} \quad (1)$$

where S represents the AI-Based Fake News Detection and Analysis System.

B. News Classification Model

A news article is classified based on extracted textual features and prediction scores generated by the trained model.

The classification function is defined as:

$$C_n = M(F_n) \quad (2)$$

Where

C_n = Classification Result

M = Trained Classification Model

F_n = Feature Vector of News Article

The classification output is:

$$C_n = \begin{cases} 1, & \text{if News is Real} \\ 0, & \text{if News is Fake} \end{cases} \quad (3)$$

Thus,

$$C_n = 1$$

indicates Real News, while

$$C_n = 0$$

indicates Fake News.

C. Prediction Probability Model

The probability that a news article belongs to the real news category is calculated using the sigmoid function:

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

where

$P(y = 1)$ = Probability of Real News

z = Weighted Feature Sum

If

$$P(y = 1) \geq 0.5$$

then News = Real

Otherwise,

$$P(y = 1) < 0.5$$

then News = Fake.

D. Text Feature Extraction Model

After preprocessing, important textual features are extracted from news articles.

The feature vector is represented as:

$$F_n = \{f_1, f_2, f_3, \dots, f_n\} \quad (5)$$

where

f_1 = Term Frequency

f_2 = Word Embeddings

f_3 = Sentiment Score

f_n = Additional Linguistic Features

These features are used by the machine learning model for news classification.

E. System Accuracy Formula

The effectiveness of the proposed system is measured using classification accuracy.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (6)$$

where

TP = True Positives

TN = True Negatives

FP = False Positives

FN = False Negatives

Example

If

$TP = 450$

$TN = 430$

$FP = 35$

$FN = 25$

Then

$$Accuracy = \frac{450 + 430}{450 + 430 + 35 + 25} \times 100$$

$$Accuracy = \frac{880}{940} \times 100$$

$$Accuracy = 93.62\%$$

F. Precision Formula

Precision measures the correctness of positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (7)$$

Example:

$$Precision = \frac{450}{450 + 35}$$

$$Precision = 0.928$$

$$Precision = 92.8\%$$

G. Recall Formula

Recall measures the ability of the system to identify fake news correctly.

$$Recall = \frac{TP}{TP + FN} \quad (8)$$

Example:

$$Recall = \frac{450}{450 + 25}$$

$$Recall = 0.947$$

$$Recall = 94.7\%$$

The obtained accuracy, precision, and recall values demonstrate the effectiveness and reliability of the proposed AI-Based Fake News Detection and Analysis System in identifying and classifying fake news articles.

V. SYSTEM IMPLEMENTATION

The implementation of the proposed AI-Based News Analysis and Credibility Verification System focuses on automating the process of analyzing and verifying online news content using artificial intelligence and natural language processing techniques. The system is developed using a ReactJS frontend and a Python FastAPI backend connected through RESTful APIs [1],[2],[5]. Users submit news headlines or article text through the frontend interface, and the backend processes the request through multiple analysis modules including text preprocessing, genre detection, evidence collection, and verdict generation.

The frontend layer is implemented using ReactJS, Vite, Tailwind CSS, React Router DOM, and Axios for secure API communication [1]. The backend is developed using FastAPI framework and transformer-based NLP models for semantic analysis and credibility evaluation [2]-[5]. The system collects references from Google News RSS feeds and Reddit public

APIs, then compares them with the submitted claim using Sentence-BERT semantic similarity and Natural Language Inference models. Multiple credibility signals such as fake-news classification, sensational language detection, sentiment analysis, and trusted source verification are combined to improve prediction accuracy.

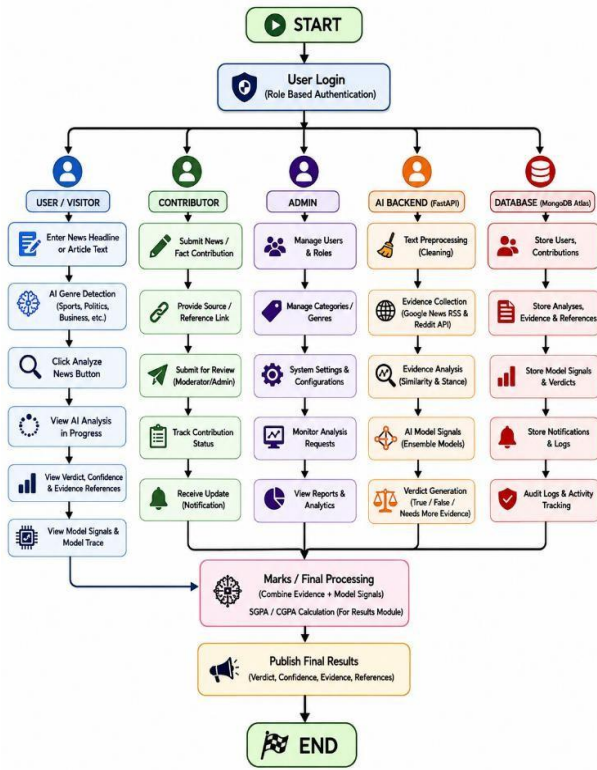


Fig. 2. Overall Flow of Fake News Detection System

After evidence analysis and credibility evaluation, the system generates final verdicts such as Likely True, Likely False, or Needs More Evidence along with confidence scores, evidence references, and model explanations. MongoDB Atlas is used for storing analysis logs, references, user contributions, and verdict data. The modular implementation improves scalability, maintainability, and explainability while reducing manual fact-checking effort. The developed system provides a responsive user interface, efficient backend processing, and transparent AI-driven news verification for digital media platforms.

The system also incorporates explainable AI mechanisms that allow users to understand how the final verdict is generated [11],[12],[13]. Instead of providing only a prediction result, the platform displays supporting evidence, semantic similarity scores, stance classifications, and credibility signal explanations. This improves transparency and helps users evaluate the reliability of the analyzed news content more effectively.

To improve performance and scalability, asynchronous processing techniques are implemented in the backend for collecting online references from multiple external sources simultaneously. Lazy loading and caching mechanisms are also used for transformer models to

reduce server startup time and optimize memory usage during analysis operations [3],[4]. These optimizations improve response speed and ensure stable backend performance during multiple user requests.

The developed implementation demonstrates how artificial intelligence, natural language processing, and evidence-based reasoning can be combined into a centralized platform for automated fake news detection and digital media verification.

VI. SYSTEM TESTING AND RESULTS

System testing was conducted to evaluate the functionality, accuracy, performance, and reliability of the proposed AI-Based News Analysis and Credibility Verification System.

Multiple testing methodologies including Unit Testing, Integration Testing, System Testing, and User Acceptance Testing were performed to validate all modules and workflows.

Unit testing verified individual modules such as text preprocessing, genre detection, evidence collection, semantic similarity computation, stance detection, and verdict generation.

Integration testing ensured proper communication between frontend components, backend APIs, transformer models, and external reference sources.

System testing validated the complete workflow of the application from user input submission to final verdict generation. The frontend successfully displayed detected genres, credibility scores, supporting references, and AI-generated explanations.

The evidence collection module successfully retrieved references from Google News RSS feeds and Reddit public APIs. Semantic similarity analysis effectively compared claims with external references using SBERT embeddings [4].

The stance detection module correctly classified evidence as supporting, contradicting, or neutral in most test cases [6],[7],[11],[13]. The model ensemble approach improved overall reliability by combining multiple AI signals instead of relying on a single model [7]-[13].

User Acceptance Testing confirmed that the system provided a responsive interface, accurate credibility analysis, real-time evidence visualization, and explainable AI outputs.

The final testing results demonstrated that the proposed system effectively reduces misinformation analysis effort, improves automated fact-checking efficiency, and provides transparent credibility evaluation for digital news content.

VII. CONCLUSION AND FUTURE SCOPE

The proposed AI-Based News Analysis and Credibility Verification System successfully automates the process of analyzing and verifying digital news content using artificial intelligence and machine learning techniques [6]-[13].

The system integrates genre detection, semantic similarity analysis, evidence collection, stance detection, and credibility signal evaluation into a centralized web-based platform. The developed solution provides reliable verdicts along with supporting references, confidence scores, and explainable AI outputs.

The frontend developed using ReactJS and Vite provides an interactive and responsive user experience, while the FastAPI backend efficiently handles NLP processing and asynchronous evidence collection. The integration of transformer-based models significantly improves semantic understanding and credibility analysis.

The proposed system helps users identify misleading information, improves awareness regarding misinformation, and contributes toward responsible digital media consumption.

In future, the system can be enhanced using advanced deep learning models, multilingual news verification, image and video fact-checking, social media trend analysis, real-time news monitoring, blockchain-based verification systems, and cloud deployment [8]-[13] for large-scale scalability. Additional features such as mobile application support, AI-generated explanation summaries, voice-based news analysis, and integration with trusted fact-checking organizations can further improve the effectiveness and usability of the platform.

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